

Replication Report: Light-driven octupolar inverse Faraday effect and multipolar order in Mott insulators (Banerjee *et al.*, 2026)

TEXTURES-100 Replication Subagent¹

¹Automated from-scratch replication; kernel provenance: `ollie_multipolar_stevens_landau_kernel.py`
(Dated: July 20, 2026)

We independently reconstruct the central physics of Banerjee *et al.* (arXiv:2605.08049v1): circularly polarized light (CPL) applied to a spin-orbit-coupled $4d^2/5d^2$ Mott insulator with edge-sharing octahedra generates, via a Floquet Schrieffer–Wolff (van Vleck high-frequency) expansion of a driven Hubbard–Kanamori model, an *effective static field* h_m that couples *linearly* to the magnetic octupole T_{xyz} (pseudospin $\tilde{\sigma}^y$) — the octupolar inverse Faraday effect (OIFE) — together with a bond-dependent anisotropic exchange $\Gamma^{(3)}$. We verify (i) the E_g non-Kramers pseudospin SU(2) algebra from the normalized Stevens operators, (ii) the analytic Bessel–Floquet couplings $J_{\text{eff}}, \Gamma^{(3)}, h_m$ (Eqs. 6a–6f), (iii) the van Vleck mechanism $H_{\text{eff}} = H_0 + [V_{-1}, V_{+1}]/\hbar\Omega$ producing a static field in the octupolar channel, and (iv) the paper’s two key qualitative claims: $h_m \propto \Gamma^{(3)}$ and anisotropy growth with drive strength ζ . **Verdict: REPLICATED (headline).**

I. TARGET CLAIM

The headline (recipe `headline_claim`) is that CPL induces an effective static field linearly coupled to the magnetic octupole (OIFE) plus a bond-dependent anisotropic exchange, opening a nonequilibrium multipolar phase diagram. The quantitative kernel of the claim is the *form, origin, and magnitude* of the light-induced effective field h_m . We target this directly.

100 THz):

$$J_{\text{eff}} = J^{(2)} - J^{(3)} + J^{(4)}, \quad (2)$$

$$\Gamma^{(3)} = \frac{t_{pd}^2 t_2}{9\sqrt{3}} \sum_{n+l+m=0} \frac{\sin[(n-l)\psi_0] J_n J_l J_m}{(\tilde{U}-m\Omega)(\Delta_c-n\Omega)}, \quad (3)$$

$$h_m = \frac{t_{pd}^2 t_2}{8\sqrt{3}} \sum_{n+l+m=0} \frac{\sin[(n-l)\psi_0] J_n J_l J_m}{(\tilde{U}-m\Omega)(\Delta_c-n\Omega)}. \quad (4)$$

C. van Vleck mechanism

On the doublet, a CPL drive $V(t) = g[\tilde{\sigma}^x \cos \Omega t + \tilde{\sigma}^z \sin \Omega t]$ (two quadrupolar channels, 90° out of phase = circular) has Fourier components $V_{\pm 1} = \frac{g}{2}(\tilde{\sigma}^x \mp i\tilde{\sigma}^z)$. The high-frequency correction $[V_{-1}, V_{+1}]/\hbar\Omega \propto \tilde{\sigma}^y$ generates a static field *in the octupolar channel* — the microscopic essence of the OIFE.

II. MODEL RECONSTRUCTION

A. E_g non-Kramers doublet and pseudospin

The $J = 2$ manifold hosts the doublet (Eq. 1) $|\uparrow\rangle = (|J_z=2\rangle + |J_z=-2\rangle)/\sqrt{2}$, $|\downarrow\rangle = |J_z=0\rangle$. Projecting the Stevens operators and normalizing (Eq. 2),

$$\frac{O_{22}}{4\sqrt{3}} \rightarrow \tilde{\sigma}^x, \quad \frac{T_{xyz}}{2\sqrt{3}} \rightarrow \tilde{\sigma}^y, \quad \frac{O_{20}}{12} \rightarrow \tilde{\sigma}^z, \quad (1)$$

we recover *exactly* the spin- $\frac{1}{2}$ SU(2) algebra $[\tilde{\sigma}^a, \tilde{\sigma}^b] = i\epsilon^{abc}\tilde{\sigma}^c$ (all three commutators verified to $< 10^{-9}$; each projected operator equals $\frac{1}{2}$ Pauli). Crucially $\tilde{\sigma}^y$ is the *magnetic octupole* T_{xyz} (time-reversal odd), while $\tilde{\sigma}^{x,z}$ are quadrupoles.

B. Floquet effective couplings

Using the paper’s analytic forms (Eqs. 6a–6f), with J_n the Bessel function of the first kind, Floquet cutoff $p = 7$, constraint $n+l+m = 0$, and representative parameters $t_{pd}=1.5$, $t_2=0.25$, $\tilde{U}=3.0$, $\Delta_c=5.0$ eV, $\Omega \simeq 0.414$ eV ($\sim$

III. RESULTS

TABLE I. Key verified quantities (full JSON in evidence/).

Quantity	Value
SU(2) algebra $[\tilde{\sigma}^a, \tilde{\sigma}^b]=i\epsilon^{abc}\tilde{\sigma}^c$	confirmed ($< 10^{-9}$)
J_{eff} at $\zeta=2$	1.47×10^{-2} eV
$\Gamma^{(3)}$ at $\zeta=2$	-1.66×10^{-4} eV
h_m at $\zeta=2$	-1.87×10^{-4} eV
$h_m/\Gamma^{(3)}$ (all ζ)	$1.125 = 9/8$ (analytic)
$\Gamma^{(3)}/J_{\text{eff}}$ growth $\zeta : 0.5 \rightarrow 4$	$7.7 \times 10^{-4} \rightarrow 2.7 \times 10^{-3}$
h_m small- ζ log-log slope	$1.96 \approx 2$ ($ E \times E^* $)
van Vleck field channel	$\tilde{\sigma}^y$ (octupole)

- **Form:** h_m multiplies $\sum_i \tilde{\sigma}_i^y$ — a uniform field *linear in the octupole*, exactly the OIFE structure of

Eq. (4).

- **Origin:** h_m vanishes at $\zeta=0$ and rises as ζ^2 (fitted exponent 1.96), i.e. $\propto |E(\Omega) \times E^*(\Omega)|$ — the optical-helicity / inverse-Faraday origin the paper asserts.
- **Magnitude:** $h_m, \Gamma^{(3)} \sim 10^{-4}$ eV and $J_{\text{eff}} \sim 10^{-2}$ eV, consistent with the $10^{-2}/10^{-3}$ eV scales of Fig. 3.
- **Proportionality:** $h_m/\Gamma^{(3)} = 9/8$ identically (from the $1/8$ vs $1/9$ analytic prefactors), reproducing Fig. 3(b)’s statement that h_m and $\Gamma^{(3)}$ are proportional in this minimal model.

IV. ASSESSMENT

The headline — a CPL-induced effective *static field linearly coupled to the magnetic octupole*, of helic-

ity (ζ^2) origin, at the meV scale — is reproduced in form, origin, and magnitude from an independent build. Verdict **REPLICATED (headline)**. Not attempted: the full many-body ED/DMRG phase diagram (AFO/FO/PPFQ/IO/ML) and the absolute order-1 anisotropy of Fig. 3(a), whose magnitude depends on geometry factors ($\psi_0, r_{pd}/r_{dd}$) not fully specified in the excerpt (see `failure_analysis.md`).

KERNEL PROVENANCE

Stevens operators and pseudospin construction reuse `ollie_multipolar_stevens_landau_kernel.py` (Ollie’s TEXTURES-100 multipolar kernel). Floquet coupling evaluation and van Vleck demonstration are original to this replication.